

TECHNICAL OVERVIEW

Ouicv

AI-Powered CV Optimization & ATS Compatibility
Platform, built for the French & European job market

ouicv.com

NEXT.JS · REACT · TYPESCRIPT · POSTGRESQL · CLERK · STRIPE · GROQ & GEMINI

1 What This Project Is

OuiCV is a full-stack SaaS web application that helps job seekers optimize their CVs to pass through **Applicant Tracking Systems (ATS)** — the automated software most companies use to filter resumes before a human ever sees them. A candidate uploads their CV (or pastes their profile), optionally provides a target job posting (by URL or pasted text), and the platform:

- Scores the CV against ATS-compatibility criteria (0–100)
- Identifies missing and matched keywords relevant to the candidate's professional domain
- Flags structural/formatting issues that confuse ATS parsers
- Uses AI to generate an optimized, rewritten version of the CV
- Renders the optimized CV into professionally designed, downloadable PDF templates
- Recommends real, live job postings matching the candidate's profile

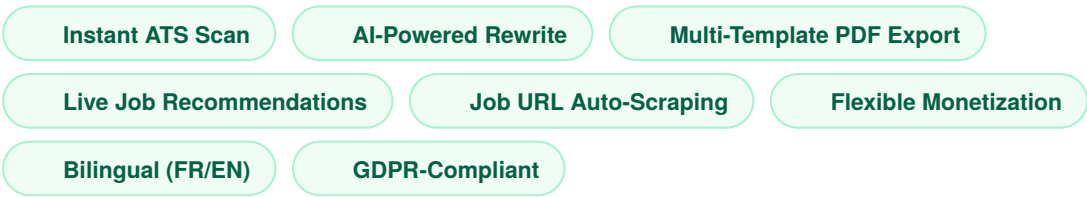
The problem it solves: most job seekers have no visibility into why their CV gets silently rejected by automated hiring software — a resume that looks perfect to a human can still fail an ATS parser due to poor formatting, missing keywords, non-standard headers, or unparseable design choices. OuiCV gives candidates the same lens recruiting software uses, then helps them fix it end to end.

2 Tech Stack

Layer	Technology
Framework	Next.js 16 (App Router), React 19, TypeScript
Styling	Tailwind CSS v4
Database	PostgreSQL (Neon, serverless, connection-pooled) via Drizzle ORM
Authentication	Clerk (email/password + Google, Facebook, LinkedIn social login)
Payments	Stripe (one-time credit packs + monthly Pro subscription)
AI / LLM	Groq (primary) + Google Gemini (automatic fallback)
File Uploads	UploadThing
File Parsing	pdf2json (PDF text extraction), mammoth (DOCX)
PDF Generation	Puppeteer + @sparticuz/chromium, pooled/reused browser instance
Rate Limiting	Upstash Redis
Internationalization	next-intl (French default, English secondary)
Job Data	France Travail (Pôle Emploi) official API + Adzuna API
Error Monitoring	Sentry
Analytics	Google Analytics 4 with Consent Mode v2

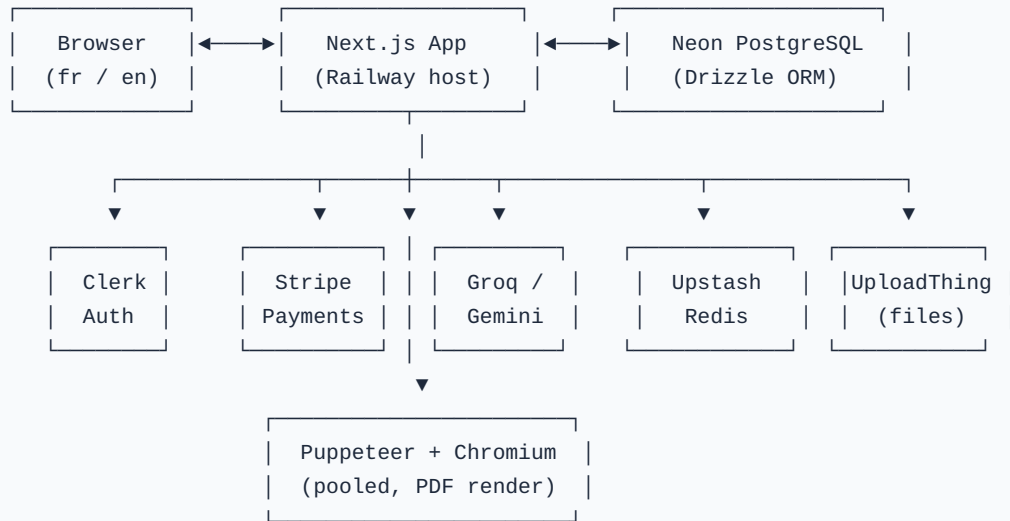
Hosting	Railway
Testing	Jest (unit tests)
CI/CD	GitHub Actions

3 Core Features



- ATS Scan** — Full breakdown across keyword match, format, experience, education, skills, and readability.
- AI-Powered CV Rewrite** — Domain-aware keyword injection across 10 professional domains (IT, healthcare, legal, engineering, trades, education, business, services, administrative, finance).
- Job Recommendations** — Live matching postings pulled from France Travail and Adzuna based on the candidate's optimized profile.
- Job URL Scraping** — Paste a job posting URL and the description is extracted automatically, with a graceful fallback to manual paste if a site blocks automated access.

4 Architecture Overview



External data: France Travail API · Adzuna API
Observability: Sentry · Google Analytics 4 (Consent Mode v2)

5 Notable Engineering Problems Solved

A snapshot of real production challenges identified and resolved — included to demonstrate the debugging and system-design work behind this project, not just the feature list.

AI reliability under free-tier constraints — Diagnosed a reasoning-model token-budget miscalculation (confusing a per-minute rate limit with a per-call output cap) causing silent, incomplete AI responses; fixed with correct token budgeting, retry-with-backoff on transient errors, and a proper primary/fallback provider chain.

Silent data-corruption bug — Found a failed AI extraction call being swallowed and saved as a "completed" analysis with blank CV fields; fixed by aborting and refunding the user's scan attempt instead of silently persisting broken data.

IDOR vulnerability — Discovered that shared CV result links exposed any user's private CV data to anyone holding the URL, with no ownership check; fixed with a consistent owner/guest-session verification layer across all CV view routes.

PDF generation memory leak → server crash — Diagnosed a production out-of-memory crash caused by an always-on headless browser instance; resolved with a pooled, concurrency-limited browser plus an idle-timeout auto-close.

ATS scoring integrity bug — Found a hardcoded fallback scoring path that silently returned a fixed score regardless of actual CV content; replaced with an honest "not yet scored" state.

Cross-browser rendering bug — Root-caused an Android-only crash to Chrome's automatic translation feature mutating the DOM behind React's back; fixed via correct `lang` attribute handling and `notranslate` directives.

Tiered rate-limiting system — Designed a multi-tier (anonymous / free / one-time / subscription) rate-limiting system with hourly and daily caps, a one-time login bonus mechanism, and automatic refunds on failed attempts.

6 Project Structure

```
├─ app/
│   ├─ [locale]/                # Localized pages (fr / en)
│   │   ├─ dashboard/
│   │   ├─ results/[analysisId]/
│   │   ├─ templates/[analysisId]/
│   │   └─ sign-in/, sign-up/
│   ├─ api/
│   │   ├─ analyze-cv/
│   │   ├─ generate-pdf/
│   │   ├─ generate-templates/
│   │   ├─ job-recommendations/
│   │   ├─ stripe/
│   │   ├─ webhooks/
│   │   └─ admin/cleanup-deleted-users/
│   ├─ robots.ts
│   └─ sitemap.ts
├─ components/
│   ├─ home/                    # Upload & job-input flow
│   ├─ results/                 # ATS score UI, job recommendations
│   ├─ templates/              # CV renderer, editor, template styles
│   └─ settings/
├─ lib/
│   ├─ ai/                      # LLM gateway, ATS analyzer, CV parser
│   ├─ billing/                # Plan logic, credit service
│   ├─ db/                     # Drizzle schema
│   ├─ jobs/                   # France Travail / Adzuna providers
│   ├─ pdf/                    # Browser pool for PDF rendering
│   ├─ rate-limit/             # Upstash rate limiters
│   ├─ scraper/                # Job URL scraping
│   └─ utils/
├─ i18n/                       # next-intl routing & config
├─ messages/                   # fr.json / en.json translations
└─ .github/workflows/          # CI/CD pipelines
```

7 Getting Started Locally

Prerequisites

- Node.js 20+, npm
- A PostgreSQL database (Neon recommended — free tier works)
- Accounts for: Clerk, Stripe, Groq, Google AI Studio (Gemini), Upstash, UploadThing

1. Clone the repository

```
git clone https://github.com/<your-org>/cvmatch-pro-v1-fresh.git
cd cvmatch-pro-v1-fresh
```

2. Install dependencies

```
npm install --legacy-peer-deps
```

3. Configure environment variables

```
cp .env.example .env.local
```

4. Set up the database

```
npx drizzle-kit generate
npx drizzle-kit push
```

5. Run the development server

```
npm run dev
```

The app will be available at <http://localhost:3000> (redirects to </fr>).

6. Run tests

```
npm run test:run
```

8 Environment Variables

Variable	Purpose
<code>DATABASE_URL</code>	Neon PostgreSQL connection string (pooled)
<code>NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY</code> / <code>CLERK_SECRET_KEY</code>	Clerk authentication
<code>CLERK_WEBHOOK_SECRET</code>	Verifies incoming Clerk webhooks
	Stripe payments

<code>NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY</code> / <code>STRIPE_SECRET_KEY</code>	
<code>STRIPE_WEBHOOK_SECRET</code>	Verifies incoming Stripe webhooks
<code>STRIPE_ONE_TIME_PRICE_ID</code> / <code>STRIPE_SUBSCRIPTION_PRICE_ID</code>	Stripe product price IDs
<code>PRIMARY_LLM_PROVIDER/MODEL/API_KEY</code>	Primary AI provider (Groq)
<code>FALLBACK_LLM_PROVIDER/MODEL/API_KEY</code>	Fallback AI provider (Gemini)
<code>UPSTASH_REDIS_REST_URL</code> / <code>_TOKEN</code>	Redis (rate limiting)
<code>UPLOADTHING_TOKEN</code>	File upload storage
<code>FRANCE_TRAVAIL_CLIENT_ID</code> / <code>_SECRET</code>	Job recommendations (France)
<code>ADZUNA_APP_ID</code> / <code>ADZUNA_APP_KEY</code>	Job recommendations (broader coverage)
<code>TRACKING_SALT</code>	Salts anonymous/guest session hashing
<code>CRON_SECRET</code>	Authenticates the GDPR cleanup cron job
<code>RESEND_API_KEY</code> / <code>CONTACT_EMAIL</code>	Transactional/contact email
<code>NEXT_PUBLIC_HCAPTCHA_SITE_KEY</code> / <code>HCAPTCHA_SECRET_KEY</code>	Bot protection
<code>NEXT_PUBLIC_SENTRY_DSN</code> / <code>SENTRY_AUTH_TOKEN</code>	Error monitoring
<code>NEXT_PUBLIC_APP_URL</code>	Canonical app URL
<code>NEXT_PUBLIC_DEFAULT_LOCALE</code>	Default locale (fr)

9 Database

PostgreSQL hosted on **Neon** (serverless, autoscaling compute, connection-pooled via PgBouncer). Schema managed with **Drizzle ORM** — schema-as-code in `lib/db/schema.ts`.

Key tables: `users`, `cvAnalyses`, `cvTemplates`, `userTemplateUnlocks`, `creditTransactions`, `usageLogs`, `jobRecommendations`, `stripeEvents`.

```
npx drizzle-kit generate # generate a new migration
npx drizzle-kit push     # apply schema changes (dev)
npx drizzle-kit studio   # visual DB browser
```

10 Testing

Type	Tool	Command
Unit tests	Jest + Testing Library	<code>npm run test:run</code>
Coverage	Jest	<code>npm run test:coverage</code>
Type safety	TypeScript	<code>npx tsc --noEmit</code>

Linting	ESLint	<code>npm run lint</code>
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11 CI/CD Pipeline

Implemented with **GitHub Actions**, across four workflows:

Workflow	Trigger	Purpose
<code>test.yml</code>	Every push (non-main)	Fast unit test feedback loop
<code>ci.yml</code>	Push / PR	Unit tests → lint → type-check → full build verification
<code>deploy.yml</code>	Push to main	Final test + build gate before Railway's auto-deploy
<code>cleanup.yml</code>	Daily cron (2 AM UTC)	GDPR-compliant hard-delete of data older than 30 days

Railway auto-deploys from `main` once GitHub Actions confirms a clean build.

12 Third-Party Services & APIs

Service	Role
Clerk	Authentication (email/password + social login)
Stripe	Subscription and one-time payment processing
Groq	Primary LLM provider for CV analysis and generation
Google Gemini	Fallback LLM provider
Upstash Redis	Rate limiting and anti-abuse tracking
UploadThing	CV file upload storage
France Travail (Pôle Emploi)	Official French government job-listing API — OAuth2 client-credentials flow, primary source for job recommendations
Adzuna	Secondary job-listing aggregator API, broadening recommendation coverage
Sentry	Real-time error monitoring and alerting
Google Analytics 4	Traffic analytics with Consent Mode v2 (GDPR)
hCaptcha	Bot/abuse protection
Resend	Transactional email delivery

13 Security & Compliance

- **Per-user data ownership enforcement** on every CV view/edit route
- **GDPR-compliant cookie consent**, wired into Google Analytics via Consent Mode v2
- **Automated 30-day hard-delete** of soft-deleted user data via scheduled cron
- **Tiered, Redis-backed rate limiting** with bot detection and IP soft-blocking
- **No password handling in application code** — fully delegated to Clerk
- **Webhook signature verification** for Clerk and Stripe, with idempotency tracking

Built with care, iterated on relentlessly, and shipped for real users.
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